LIMITS OF MATRIX APPROXIMATION OF GROUPS IN OPERATOR NORM

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We give a negative answer to the longstanding question whether every countable group is MF. Here MF means admitting finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. Our main tool is the following one-sided compression criterion. Let G be countable, let $L \leq G$ have property (T), and let $K \trianglelefteq G$ have property (T). If K is contained in the normal subgroup generated by the commutators $[ucu^{-1}, \ell]$, where $uLu^{-1} \leq L$, $c \in C_G(L)$, and $\ell \in L$, then every homomorphism from G to an MF group kills K. For $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$, a single commutator of this form normally generates H. The group H is nontrivial, simple, and has property (T), whereas every homomorphism from H to an MF group is trivial. In particular, H is not MF. Consequently, the reduced group C^*-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF. Finally, if $f\colon G \to Q$ is surjective and $\ker f \leq \mathrm{Rad}_{\mathrm{MF}}(G)$, then $\mathrm{Rad}_{\mathrm{MF}}(G) = f^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q))$. In this case, image and inverse image identify the normal subgroups $N \trianglelefteq G$ satisfying $\ker f \leq N$ and G/N MF with the normal subgroups $P \trianglelefteq Q$ satisfying Q/P MF.

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1. Introduction

Not every countable group is MF. Our counterexample is

$$H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2)).$$

The group H is countable, nontrivial, simple, and has property (T), but every homomorphism from H to an MF group is trivial. In particular, H is not MF. Its reduced group C^*-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF.

More precisely, a countable group G is *MF* if there are positive integers d_n and maps $V_n\colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

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and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

Thus the models become multiplicative in operator norm without losing any nonidentity element. This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE].

Equivalently, G embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \longrightarrow 0\}$$

is the c_0-direct sum.

A separable C^*-algebra is called MF if it embeds as a C^*-subalgebra of a norm matrix corona. For every countable group G, the reduced group algebra $C^*_{\mathrm{r}}(G)$ is separable and has a faithful canonical trace, hence is stably finite. If $C^*_{\mathrm{r}}(G)$ is MF, restricting an MF embedding to its canonical group unitaries shows that G is MF. Therefore a non-MF group automatically gives a separable stably finite reduced group C^*-algebra that is not MF.

For a countable group G, define its MF radical by

$$\mathrm{Rad}_{\mathrm{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi \colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

Proposition 1.1 shows that G is MF precisely when $\mathrm{Rad}_{\mathrm{MF}}(G)$ is trivial. If $\mathrm{Rad}_{\mathrm{MF}}(G) = G$, every homomorphism from G to an MF group is trivial.

More generally, for $N \trianglelefteq G$ define its *MF kernel closure* by

$$\mathrm{cl}^G_{\mathrm{MF}}(N) = \bigcap \{\ker f : N \le \ker f,\ f \colon G \to M,\ M \text{ MF}\}.$$

The image of a corona homomorphism from G is countable and is itself MF; conversely, every MF target embeds in a norm matrix corona. Thus $\mathrm{Rad}_{\mathrm{MF}}(G) = \mathrm{cl}^G_{\mathrm{MF}}(1)$, and G/N is MF precisely when $\mathrm{cl}^G_{\mathrm{MF}}(N) = N$. This is a closure operation on the normal subgroups of one fixed group; it is unrelated to topological closure of classes of marked groups.

Proposition 1.1 (basic properties of the MF radical)**.** *For every countable group G, the subgroup $\mathrm{Rad}_{\mathrm{MF}}(G)$ is fully invariant, the quotient $G/\mathrm{Rad}_{\mathrm{MF}}(G)$ is MF, and*

$$G/N \text{ is MF} \iff \mathrm{cl}^G_{\mathrm{MF}}(N) = N \qquad (N \trianglelefteq G).$$

In particular, G is MF if and only if its MF radical is trivial.

Proof. An intersection of kernels is normal. If α is an endomorphism of G, $x \in \mathrm{Rad}_{\mathrm{MF}}(G)$, and π is a corona homomorphism of G, then $\pi \circ \alpha$ kills x. Hence $\pi(\alpha(x)) = 1$ for every π, which proves full invariance.

Put $R = \mathrm{Rad}_{\mathrm{MF}}(G)$ and enumerate the nonidentity elements of G/R as $(x_j)_{j \ge 1}$ and the pairs in $(G/R)^2$. For each x_j, the definition of R supplies a corona homomorphism whose value at x_j has positive distance from the identity. Every such homomorphism kills R and therefore descends to G/R. Choose coordinate unitary lifts, using polar correction as in Lemma 4.1. At stage n, take a direct sum of one coordinate from each of the first n models. Choose each coordinate far enough out that the first n multiplication defects are at most $1/n$ and the designated value at x_j retains at least half of its corona norm separation. Such coordinates exist arbitrarily far out: the defects converge to zero, while the quotient norm is a limsup. The resulting operator norm asymptotic representation detects every x_j, so its corona homomorphism is faithful on G/R. Thus G/R is MF.

Applied to G/N, the same argument says that G/N is MF exactly when the intersection of the kernels of all maps into MF groups that kill N is N, which is the asserted closure criterion. □

For a subgroup $L \leq G$, put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression–centralizer defect* by

$$\mathfrak{D}_G(L) = \langle\!\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\!\rangle_G. \tag{1}$$

Theorem A (one-sided compression criterion)**.** *Let G be countable and let $L \leq G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \mathrm{Rad}_{\mathrm{MF}}(G).$$

In particular, a nontrivial such K makes G non-MF. If G has property (T) *and $\mathfrak{D}_G(L) = G$, then $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.*

We sketch the argument. Corollary 3.4 gives $\mathfrak{D}_G(L) \leq R_{\infty\to 2}(G)$. Equivalently, if (V_n) is any operator norm asymptotic representation of G, then $\|V_n(d) - 1\|_2 \to 0$ for every $d \in \mathfrak{D}_G(L)$. To upgrade this conclusion to corona triviality on K, fix a homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ and suppose, toward a contradiction, that Θ is nontrivial on K. Let p be the image in $\mathcal{Q}_{\mathbf{d}}$ of the Kazhdan projection of K. Because $K \trianglelefteq G$, the projection p commutes with $\Theta(G)$; hence $q = 1 - p$ defines a G-invariant complementary corner. Lemma 4.1 realizes the action on this corner by an operator norm asymptotic representation (W_n) on matrix corners with identities q_n. The corner has no nonzero K-fixed vectors, so the Kazhdan inequality yields a single $s_0 \in K$ and a subsequence along which $\|W_n(s_0) - q_n\|_2$ stays bounded below by a positive constant. This contradicts $s_0 \in K \leq \mathfrak{D}_G(L)$ and the preceding convergence. Thus every corona homomorphism kills K, as required.

Theorem 2.1 gives a separate finite-dimensional vanishing result: every finite-dimensional linear representation of G over any field kills $\mathfrak{D}_G(L)$.

We next apply Theorem A to the binary Leavitt construction that produces the group H announced above.

Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$-algebra generated by s_0, s_1, t_0, t_1 subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{2}$$

For a unital ring R, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

Theorem B (the binary Leavitt group)**.** *Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. Then H is countable, nontrivial, simple, has property* (T)*, and*

$$\mathrm{Rad}_{\mathrm{MF}}(H) = H.$$

Equivalently, every homomorphism from H to an MF group is trivial. In particular, H is non-MF. Its reduced group C^*-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF.

Inside H, an explicit element τ satisfies $\tau L \tau^{-1} \leq L$ for the subgroup $L \cong \mathrm{EL}_3(R)$ in the upper left. The element $c = e_{34}(1)$ centralizes L, and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The relation $t_1(s_1 t_1)s_1 = 1$ and the elementary commutator relations show directly that d normally generates H. Since $d \in \mathfrak{D}_H(L)$ and $\mathfrak{D}_H(L)$ is normal, it follows that $\mathfrak{D}_H(L) = H$. Therefore Theorem A gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$.

If $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism and V is finite-dimensional, conjugation by $\rho(u)$ maps the commutant of $\rho(L)$ injectively into itself, hence bijectively. Thus ρ kills $\mathfrak{D}_G(L)$. Consequently, $\mathfrak{D}_G(L) = 1$ if G has a faithful finite-dimensional linear representation. The same conclusion holds if G is residually finite, because every nonidentity element then survives in a finite permutation representation. If G is amenable, its property-(T) subgroup L is finite [BHV]; then $uLu^{-1} = L$, so ucu^{-1} centralizes L for every $u \in \mathrm{Comp}_G(L)$, and again $\mathfrak{D}_G(L) = 1$.

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 3.3.

A full MF radical also lets us prescribe the quotient visible to MF.

Theorem C (prescribed quotients visible to MF)**.** *Let B be a countable group with $\mathrm{Rad}_{\mathrm{MF}}(B) = B$, and let $1 \neq d \in B$ normally generate B. Put $A = \langle d \rangle$ and, for a countable group Q, define*

$$W_Q = B *_A (Q \times A).$$

Then the map $\pi_Q\colon W_Q \to Q$ which kills B and projects the second vertex group onto Q is a split epimorphism, and

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The group W_Q is non-MF. If Q is MF, then

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\!\langle d \rangle\!\rangle_{W_Q}.$$

Moreover, precomposition with π_Q induces a bijection

$$\mathrm{Hom}(Q, M) \longrightarrow \mathrm{Hom}(W_Q, M)$$

for every MF group M.

Relation to prior work. The operator-algebraic lineage of the problem predates the term *MF group*. In a 1994 Oberwolfach abstract, Blackadar and Kirchberg suggested that every separable nuclear stably finite C^*-algebra might be NF [BK94]. Their 1997 paper introduced MF C^*-algebras; for separable nuclear algebras, the MF and NF properties are equivalent [BK]. They also observed that no separable stably finite C^*-algebra was known not to be MF. The broader question whether every separable stably finite C^*-algebra is MF became known as the Blackadar–Kirchberg MF problem [RS].

Carrión–Dadarlat–Eckhardt later introduced the precise group property used here [CDE]. Subsequent approximation literature attributed the question whether every countable group is MF to Kirchberg [LO, Th]. The negative solution of the Connes embedding problem also produced stably finite non-MF C^*-algebras in general [JNVWY]. Theorem B provides such an example among reduced group C^*-algebras and answers the operator norm group approximation question negatively. Korchagin subsequently developed permanence and local approximation results for MF groups [Kor]. Shulman proves MF permanence for doubles $G *_H G$ of MF groups. Separately, Shulman proves MF full group C^*-algebra results for several classes of amalgamated and semidirect products [Sh]. The operator norm case remained open after failure of approximation had been established for finite Schatten norms [DGLT, LO, Th]. Bachner–Dogon–Lubotzky study stability between the operator and Hilbert–Schmidt norms as a possible source of non-MF groups [BDL]. Here operator norm control implies that coordinatewise conjugation defines a representation in a second norm matrix corona. The Kazhdan projection of this conjugation representation makes the Hilbert–Schmidt asymptotic commutant invariant under the one-sided compressor. Theorem 4.2 then proves that a normal property-(T) subgroup contained in the defect lies in the MF radical.

Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. Their defining relations supply the internal compression used here. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser classified the subgroups of GL_n over an exchange ring that are normalized by EL_n [Pre]. In the present case, every nontrivial normal subgroup of $\mathrm{EL}_{12}(R)$ contains some $e_{ij}(x)$ with $i \neq j$ and $x \neq 0$ (Proposition 6.3). Ershov–Jaikin-Zapirain give property (T) for $\mathrm{EL}_n(R)$ whenever $n \geq 3$ and R is a finitely generated unital associative ring [EJZ, Theorem 1.1]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

2. One-sided compression in finite dimension

Theorem 2.1 (finite-dimensional commutant rigidity)**.** *Let G be a group, let $L \le G$, and suppose $u \in G$ satisfies $uLu^{-1} \le L$. If k is a field, V is a finite-dimensional k-vector space, and $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where $\rho(L)'$ is the commutant of $\rho(L)$ in $\mathrm{End}_k(V)$. Consequently, if $c \in C_G(L)$, then

$$\rho([ucu^{-1}, \ell]) = 1 \qquad (\ell \in L).$$

Proof. Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned}\rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u)\\ &= \rho(u)^{-1}x\rho(h')\rho(u)\\ &= \rho(u)^{-1}x\rho(u)\rho(h).\end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and C is finite-dimensional. Its restriction to C is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. Therefore $\rho([ucu^{-1}, \ell]) = 1$ for every $\ell \in L$. □

The finite-dimensional hypothesis is essential here. On an infinite-dimensional space an injective endomorphism of the commutant need not be surjective. In Section 3 the Kazhdan projection places the analogous endomorphism inside a norm matrix corona, where stable finiteness supplies the missing surjectivity for projections onto fixed vectors.

3. Kazhdan transport in normalized Hilbert–Schmidt norm

For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator norm asymptotic representation* of a countable group G is a sequence of maps $V_n\colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of G. Define

$$R_{\infty\to 2}(G) = \bigcap_V K_2(V), \tag{3}$$

where V ranges over all operator norm asymptotic representations of G.

For $L \le G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\mathrm{Ad}(V(g))$, of the bounded matrix sequences.

Lemma 3.1. *For every sequence (m_n), the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.

Proof. Suppose that $v^*v = 1$ in the corona, and let (x_n) be a bounded lift of v. Then $x_n^* x_n \to 1$ in norm. For all sufficiently large n, the matrix x_n is invertible, and the unitary $x_n(x_n^* x_n)^{-1/2}$ differs from x_n by $o(1)$. Thus v is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. If A is a stably finite unital C^*-algebra, equivalent projections $p \le q$ satisfy $p = q$. Indeed, if $w^*w = q$ and $ww^* = p$, then $w \in qAq$ is an isometry in the corner. The corner is finite because $w + (1-q)$ is an isometry in A. Consequently $ww^* = q$, and hence $p = q$. □

Lemma 3.2 (one-sided order for the Kazhdan projection)**.** *Let L have property* (T)*, let B be a unital C^*-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

then

$$U^*PU \le P.$$

Proof. Represent B faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection P is the orthogonal projection onto $\mathrm{Fix}\,\pi(L)$. If ξ is L-fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*\big(U\pi(\ell)U^*\big)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^*\,\mathrm{Fix}\,\pi(L) \subseteq \mathrm{Fix}\,\pi(L)$, so the range projection U^*PU is dominated by P in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in B. □

Theorem 3.3 (one-sided Kazhdan transport)**.** *Let $L \leq G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let (V_n) be an operator norm asymptotic representation of G. Then*

$$\operatorname{Ad}(V(u))(\mathcal{C}_2(V,L)) = \mathcal{C}_2(V,L).$$

Proof. Fix $x = (x_n) \in \mathcal{C}_2(V,L)$ and let ω be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation σ of G on $\mathcal{K}_\omega$. Indeed, for unitaries A, B one has

$$\|\operatorname{Ad}(A) - \operatorname{Ad}(B)\| \leq 2\,\|A - B\|\,,$$

so the operator norm multiplicative defects of (V_n) also make the conjugation actions multiplicative modulo c_0. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\widetilde{\sigma}(g)$ be the class of the coordinate operators $\operatorname{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C}))/\bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes d_n^2. The preceding estimate shows directly that $g \mapsto \widetilde{\sigma}(g)$ is an exact homomorphism $G \to \mathcal{U}(\mathcal{B})$. Its restriction to L therefore extends to a $*$-homomorphism $C^*_{\max}(L) \to \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\operatorname{Fix}\sigma(L)$. This representation of $\mathcal{B}$ need not be faithful; it is used only after the following projection identity has been proved inside $\mathcal{B}$. Since $uLu^{-1} \leq L$, the unitary $U = \widetilde{\sigma}(u)$ satisfies the hypothesis of Lemma 3.2, and hence $U^*PU \leq P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 3.1 therefore gives $U^*PU = P$, so U commutes with P. Hence both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since ω was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus $\operatorname{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V,L)$, which proves the equality. ∎

Corollary 3.4. *Let $L \leq G$ have property* (T)*, let $uLu^{-1} \leq L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G) \qquad (\ell \in L).$$

Proof. Let (V_n) be an operator norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V,L)$. By Theorem 3.3, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\left\| V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1}) \right\| \longrightarrow 0.$$

Therefore

$$\left\| V_n([ucu^{-1}, \ell]) - 1 \right\|_2 \longrightarrow 0 \qquad (\ell \in L).$$

Intersecting over (V_n) proves the claim. □

4. Normal Kazhdan subgroups and the MF radical

Let $K \trianglelefteq G$ have property (T), and let p be the image of its Kazhdan projection under a corona representation of G. Normality of K implies that p commutes with the image of G. Hence $q = 1 - p$ also commutes with that image, and the corner $q\mathcal{Q}_{\mathbf{d}}q$ has no nonzero K-fixed vectors.

Lemma 4.1 (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$ and an operator norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(q_n M_{d_n}(\mathbb{C}) q_n)$$

whose corona class is the corner representation $g \mapsto q\rho(g)$.

Proof. Lift q to projections q_n, and lift each $\rho(g)$ to unitaries $U_n(g)$. The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \qquad (g \in G).$$

Consequently $q_n U_n(g) q_n$, viewed on $q_n \mathbb{C}^{d_n}$, has both unitarity defects converging to zero. Its polar correction $W_n(g)$ is unitary and satisfies

$$\|W_n(g) - q_n U_n(g) q_n\| \longrightarrow 0.$$

The homomorphism relation for ρ now makes (W_n) an operator norm asymptotic representation. Since $q \neq 0$, infinitely many q_n are nonzero; retain those coordinates. □

Theorem 4.2 (normal Kazhdan radical theorem)**.** *Let G be countable, let $D \leq R_{\infty\to 2}(G)$, and let $K \trianglelefteq G$ have property* (T) *with $K \leq D$. Then*

$$K \leq \mathrm{Rad}_{\mathrm{MF}}(G).$$

Proof. Suppose that a homomorphism $\Theta\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on K. Replace G by $\Theta(G)$, replace K by $\Theta(K)$, and denote the resulting inclusion into the corona again by Θ. The new K is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. Every one of its elements remains in the universal Hilbert–Schmidt kernel, because any operator norm asymptotic representation of $\Theta(G)$, precomposed with the quotient map from the original group, is such a representation of the original group.

Let $p_K \in C^*_{\max}(K)$ be the Kazhdan projection and let p be its image in $\mathcal{Q}_{\mathbf{d}}$. Under any faithful representation of the corona, p is the projection onto the K-fixed vectors. Normality of K gives

$$\Theta(g) p \Theta(g)^* = p \qquad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges $\Theta(g)\,\mathrm{Fix}\,\Theta(K)$ and $\mathrm{Fix}\,\Theta(K)$; these agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection q is nonzero: if $p = 1$, then every vector is fixed by K, so every element of K has trivial corona image. Lemma 4.1 gives an operator norm asymptotic representation (W_n) on nonzero corners $q_n M_{d_n}(\mathbb{C}) q_n$.

Choose a finite symmetric Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. In the corner $q\mathcal{Q}_{\mathbf{d}}q$, the Kazhdan projection is zero. Hence the positive element

$$b = \frac{1}{|S|}\sum_{s\in S}(q\Theta(s)q - q)^*(q\Theta(s)q - q)$$

satisfies

$$b \geq \frac{\kappa^2}{|S|}\, q.$$

Indeed, in every representation of the corner there are no K-fixed vectors, so the defining Kazhdan inequality gives this operator inequality.

The coordinate elements

$$b_n = \frac{1}{|S|}\sum_{s\in S}(W_n(s) - q_n)^*(W_n(s) - q_n)$$

represent b. Taking the normalized trace on $q_n M_{d_n}(\mathbb{C}) q_n$ yields

$$\frac{1}{|S|}\sum_{s\in S}\|W_n(s) - q_n\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

Here the Hilbert–Schmidt norms use the normalized trace of the corner. The order inequality for b says that the negative part of $b_n - (\kappa^2/|S|)q_n$ converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed $s_0 \in S$ therefore stays a positive Hilbert–Schmidt distance from the corner identity. This contradicts $s_0 \in K \leq D \leq R_{\infty\to 2}(G)$. Thus every corona homomorphism kills K, as required. □

Proof of Theorem A. For every $u \in \mathrm{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 3.4 gives

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G).$$

Normality of $R_{\infty\to 2}(G)$ therefore gives

$$\mathfrak{D}_G(L) \leq R_{\infty\to 2}(G).$$

Theorem 4.2 now gives $K \leq \mathrm{Rad}_{\mathrm{MF}}(G)$. Taking a nontrivial K gives the non-MF conclusion. Taking $K = G = \mathfrak{D}_G(L)$ gives $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.

For a finite-dimensional linear representation over an arbitrary field, Theorem 2.1 kills every generator of $\mathfrak{D}_G(L)$. Its kernel is normal, so it contains $\mathfrak{D}_G(L)$. □

Proposition 4.3 (functoriality and normal generation)**.** *Let $L \le G$ and put $D = \mathfrak{D}_G(L)$.*

If $f\colon G \to Q$ is a homomorphism, then

$$f(D) \le \mathfrak{D}_{f(G)}(f(L)). \tag{4}$$

If $S \le G$ is a nontrivial simple subgroup and $D \cap S \ne 1$, then $S \le D$. If in addition the normal closure of S in G is all of G, then $D = G$. If moreover G is countable and both L and G have property (T)*, then*

$$\mathrm{Rad}_{\mathrm{MF}}(G) = G.$$

Alternatively, suppose that f is onto, $S \le D$ is simple, $f(S) \ne 1$, and $f(S)$ normally generates Q. Then

$$\mathfrak{D}_Q(f(L)) = Q.$$

If, in addition, G is countable and both L and G have property (T)*, then*

$$\mathrm{Rad}_{\mathrm{MF}}(Q) = Q.$$

Proof. If $u \in \mathrm{Comp}_G(L)$, then $f(u)f(L)f(u)^{-1} \le f(L)$. If $c \in C_G(L)$, then $f(c) \in C_{f(G)}(f(L))$. Therefore f takes every defining generator of D into a defining generator of $\mathfrak{D}_{f(G)}(f(L))$. This proves (4).

The subgroup D is normal in G, so $D \cap S$ is normal in S. Simplicity and $D \cap S \ne 1$ give $D \cap S = S$, hence $S \le D$. Since D is normal in G, it then contains the normal closure of S. When the normal closure of S is G, this proves $D = G$. Under the property (T) hypotheses, Theorem A with $K = G$ then gives $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.

For the image statement, (4) gives $f(S) \le \mathfrak{D}_Q(f(L))$. The latter subgroup is normal in Q, so it contains the normal closure of $f(S)$ and is therefore all of Q. Property (T) passes to quotients, so the final hypotheses give property (T) for both $f(L)$ and Q. Applying Theorem A inside Q, with $K = Q$, proves the last assertion. ☐

5. The binary Leavitt self-compression

Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0 t_0, \qquad q = s_1 t_1.$$

The relations (2) give

$$p + q = 1, \qquad t_1 q s_1 = 1. \tag{5}$$

In particular $q \ne 0$.

We use indices $0, \dots, 11$ for 12×12 matrices. We identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. On $M_3(R)$ consider the unital injective endomorphism

$$\Psi(A) = qI_3 + s_0 A t_0. \tag{6}$$

Here s_0At_0 denotes the matrix $(s_0a_{ij}t_0)_{ij}$. The relations $qs_0 = t_0q = 0$ and $t_0s_0 = 1$ show that Ψ is multiplicative. Moreover,

$$\Psi(I_3) = qI_3 + s_0I_3t_0 = qI_3 + pI_3 = (p+q)I_3 = I_3,$$

so Ψ is unital. Finally, $t_0\Psi(A)s_0 = A$, and hence Ψ is injective.

To implement Ψ by conjugation on the embedded copy of $\mathrm{GL}_3(R)$, define the following 6×6 block matrices:

$$X = \begin{pmatrix} s_0I_3 & s_1t_0I_3 \\ 0 & t_1I_3 \end{pmatrix}, \qquad Y = \begin{pmatrix} t_0I_3 & 0 \\ s_0t_1I_3 & s_1I_3 \end{pmatrix}.$$

A block multiplication using (2) gives $XY = YX = I_6$, so $Y = X^{-1}$. Put

$$\tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R). \tag{7}$$

A direct block multiplication gives

$$\tau\ \mathrm{diag}(A, I_9)\,\tau^{-1} = \mathrm{diag}(\Psi(A), I_9). \tag{8}$$

Lemma 5.1. *The matrix τ defined in (7) belongs to* $\mathrm{EL}_{12}(R)$.

Proof. Put $I = I_6$. The Whitehead factorization gives

(9)

$$\mathrm{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix}\begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix}\begin{pmatrix} I & X \\ 0 & I \end{pmatrix}\begin{pmatrix} I & 0 \\ I & I \end{pmatrix}\begin{pmatrix} I & -I \\ 0 & I \end{pmatrix}\begin{pmatrix} I & 0 \\ I & I \end{pmatrix}.$$

If $B = (b_{rs})_{r,s=0}^5 \in M_6(R)$, then

$$\begin{pmatrix} I & B \\ 0 & I \end{pmatrix} = \prod_{r,s=0}^{5} e_{r,6+s}(b_{rs}), \qquad \begin{pmatrix} I & 0 \\ B & I \end{pmatrix} = \prod_{r,s=0}^{5} e_{6+r,s}(b_{rs}).$$

In either product, distinct off-diagonal matrix units have product zero, so no cross-terms occur. Every factor on the right-hand side of (9) therefore belongs to $\mathrm{EL}_{12}(R)$. Since $Y = X^{-1}$, its left-hand side is τ, and the result follows. $\square$

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \le H$ denote the image of the embedding $A \mapsto \mathrm{diag}(A, I_9)$.

Proposition 5.2. *The groups H and L have property* (T)*, and $\tau \in H$ satisfies*

$$\tau L \tau^{-1} \le L. \tag{10}$$

Proof. The ring R is generated by s_0, s_1, t_0, t_1 as a unital ring. The theorem of Ershov and Jaikin-Zapirain therefore gives property (T) for $\mathrm{EL}_{12}(R)$ and $\mathrm{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 5.1 gives $\tau \in H$.

For $i \ne j$ and $a \in R$, equations (8) and (6) give

$$\tau e_{ij}(a)\tau^{-1} = e_{ij}(s_0at_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of L. These matrices generate L, so (10) follows. $\square$

6. Simplicity and the full MF radical

We first prove that H is simple. The argument uses the following consequence of pure infiniteness. For every $0 \neq x \in R$, there are $a, b \in R$ such that

$$axb = 1. \tag{11}$$

Indeed, R is purely infinite simple, so this follows from [AA, Proposition 10(v)] by taking the target element to be 1.

Lemma 6.1 (Normal generation by an elementary transvection)**.** *Let S be a unital ring, let $n \geq 3$, and let $N \trianglelefteq \mathrm{EL}_n(S)$. Suppose that $e_{ij}(x) \in N$ and that $axb = 1$ for some $a, b \in S$. Then $N = \mathrm{EL}_n(S)$.*

Proof. Choose k different from i and j. Normality of N and the elementary commutator relations give

$$[e_{ij}(x), e_{jk}(b)] = e_{ik}(xb), \qquad [e_{ji}(a), e_{ik}(xb)] = e_{jk}(1),$$

so $e_{jk}(1) \in N$.

Products of the generalized permutation matrices

$$w_{uv} = e_{uv}(1)e_{vu}(-1)e_{uv}(1)$$

conjugate $e_{jk}(1)$ to $e_{uv}(1)$ or its inverse for any $u \neq v$. Normality of N, and taking inverses if necessary, therefore gives $e_{uv}(1) \in N$ for every $u \neq v$. Given $r \in S$ and $u \neq v$, choose h different from u and v. Then

$$[e_{uh}(r), e_{hv}(1)] = e_{uv}(r) \in N.$$

Thus N contains every elementary generator of $\mathrm{EL}_n(S)$. $\square$

Lemma 6.2 (Coefficient separation)**.** *If $r, s \in R$ are nonzero, then there exist $a, b \in R$ such that*

$$arb = 0, \qquad bsar \neq 0. \tag{12}$$

Proof. By (11), choose $u, v, c, d \in R$ such that

$$urv = 1, \qquad csd = 1,$$

and set

$$a = dt_1u, \qquad b = vs_0c.$$

The Leavitt relation $t_1s_0 = 0$ gives

$$arb = dt_1(urv)s_0c = dt_1s_0c = 0.$$

On the other hand,

$$bsar = vs_0(csd)t_1ur = vs_0t_1ur.$$

If this element were zero, then

$$0 = (ur)(vs_0t_1ur)v = (urv)s_0t_1(urv) = s_0t_1,$$

contrary to $t_0(s_0t_1)s_1 = 1$. $\square$

Proposition 6.3. *The group $H = \mathrm{EL}_{12}(R)$ is nontrivial and simple.*

Proof. Let $N \trianglelefteq H$ be nontrivial. By (11) and Lemma 6.1, it is enough to find $i \neq j$ and $0 \neq x \in R$ such that $e_{ij}(x) \in N$.
The diagonal case. Choose $1 \neq g \in N$ and write $g = \mathrm{diag}(\lambda_0, \ldots, \lambda_{11})$. If g commutes with every $e_{ij}(a)$, then the relations with $a = 1$ give $\lambda_i = \lambda_j$ for all $i \neq j$. Hence $g = \lambda I_{12}$ for some unit $\lambda \in R$. The relations for arbitrary $a \in R$ then give $\lambda a = a\lambda$, so $\lambda \in Z(R)^\times$. Since $Z(R) = \mathbb{F}_2$ by [AC, Corollary 4.3], this forces $\lambda = 1$, contrary to the choice of g.

It follows that $[g, e_{ij}(a)] \neq 1$ for some $i \neq j$ and $a \in R$. Set $x = \lambda_i a \lambda_j^{-1} - a$. Then $x \neq 0$, and normality of N gives

$$[g, e_{ij}(a)] = e_{ij}(x) \in N.$$

The nondiagonal case. Suppose now that g is not diagonal. Choose distinct i, ℓ such that $g_{\ell i} \neq 0$, and put $r = (g^{-1})_{\ell i}$.

If $r = 0$, set

$$A = gE_{i\ell}g^{-1}, \qquad B = E_{i\ell}.$$

Here $A^2 = B^2 = AB = 0$. Expanding the commutator gives

$$[ge_{i\ell}(1)g^{-1}, e_{i\ell}(1)] = 1 - BA \tag{13}$$

and its image in H/N is $[e_{i\ell}(1), e_{i\ell}(1)] = 1$. Thus $1 - BA \in N$. Its defect $-BA$ is square-zero and supported in row i. There is an m for which $g_{\ell i}(g^{-1})_{\ell m} \neq 0$: otherwise the (ℓ, ℓ) entry of $g^{-1}g = 1$ would give $g_{\ell i} = 0$. Thus the (i, m) entry of the defect in (13) is nonzero.

If $r \neq 0$, apply Lemma 6.2 to r and $s = g_{\ell i}$, obtaining a, b with $arb = 0$ and $bsar \neq 0$. Set

$$A = g(aE_{i\ell})g^{-1}, \qquad B = bE_{i\ell}.$$

Every column of AB except possibly column ℓ is zero, and

$$(AB)_{k\ell} = g_{ki}arb = 0 \qquad (0 \leq k \leq 11).$$

Thus $AB = 0$. Since $A^2 = B^2 = 0$, direct expansion gives

$$[ge_{i\ell}(a)g^{-1}, e_{i\ell}(b)] = 1 - BA. \tag{14}$$

In H/N, the left-hand side is the commutator of $e_{i\ell}(a)$ and $e_{i\ell}(b)$, hence is trivial. Therefore $1 - BA \in N$. The defect is square-zero and supported in row i, while its (i, i) entry is $-bg_{\ell i}a(g^{-1})_{\ell i} = -bsar \neq 0$.

Thus in either nondiagonal case, N contains $1 + v$ for some matrix v supported in row i, with $v^2 = 0$ and $v_{im} \neq 0$ for some m. Choose n different from both i and m (with only $n \neq i$ needed when $m = i$). Direct multiplication gives

$$[1 + v, e_{mn}(1)] = e_{in}(v_{im}) \in N.$$

Since $v_{im} \neq 0$, Lemma 6.1 gives $N = H$. Finally, $e_{01}(1) \neq 1$, so H is nontrivial. □

Proposition 6.4. *Let*

$$c = e_{34}(1), \qquad \ell = e_{12}(1), \qquad d = [\tau c \tau^{-1}, \ell]$$

in H. Then

$$c \in C_H(L), \qquad d = e_{02}(q) \neq 1, \qquad \langle\langle d \rangle\rangle_H = H.$$

Proof. For $i, j \in \{0, 1, 2\}$ with $i \neq j$ and $a \in R$, we have $j \neq 3$ and $i \neq 4$. Hence the Steinberg relations give

$$[e_{ij}(a), e_{34}(1)] = 1.$$

These matrices generate L, so $c = e_{34}(1)$ belongs to $C_H(L)$.

Direct multiplication of the matrices in (7) gives

$$\tau c \tau^{-1} = e_{01}(q) e_{34}(1). \tag{15}$$

The second factor commutes with $\ell = e_{12}(1)$. The identity $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ therefore gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$. Since $t_1 q s_1 = 1$, Lemma 6.1, applied to $e_{02}(q)$, gives $\langle\langle d \rangle\rangle_H = H$. $\square$

Proof of Theorem B. The finitely generated $\mathbb{F}_2$-algebra R is countable, and hence so is H. The ring R is finitely generated, and H has property (T) by the Ershov–Jaikin-Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (10), the centrality of c relative to L, and Proposition 6.4 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 6.4 gives $\langle\langle d \rangle\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group H has property (T), so Theorem A, applied with $K = H$, gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$. Independently of this radical calculation, Proposition 6.3 shows that H is nontrivial and simple.

If M is MF and $f : H \to M$ is a homomorphism, compose f with a faithful corona embedding of M. Since $\mathrm{Rad}_{\mathrm{MF}}(H) = H$, this composition and hence f are trivial. If H were MF, its identity homomorphism would contradict this conclusion.

Finally, $C^*_{\mathrm{r}}(H)$ is separable because H is countable. Its canonical trace is faithful, so it is stably finite. If $C^*_{\mathrm{r}}(H)$ were MF, restricting an MF embedding to the canonical unitaries would make H an MF group, a contradiction. $\square$

7. Quotients visible to MF

The MF radical is exact across every surjection whose kernel is already invisible to MF.

Proposition 7.1 (full-kernel pullback)**.** *Let $f : G \to Q$ be a surjective homomorphism of countable groups. If $\ker f \leq \mathrm{Rad}_{\mathrm{MF}}(G)$, then*

$$\mathrm{Rad}_{\mathrm{MF}}(G) = f^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

In particular, the conclusion holds whenever $\mathrm{Rad}_{\mathrm{MF}}(\ker f) = \ker f$*. More generally, for every normal subgroup* $N \trianglelefteq G$*,*

$$\mathrm{cl}^{G}_{\mathrm{MF}}(N) = f^{-1}\Big(\mathrm{cl}^{Q}_{\mathrm{MF}}(f(N))\Big).$$

Consequently,

$$G/N \text{ is MF} \iff \ker f \le N \text{ and } Q/f(N) \text{ is MF.}$$

Proof. Every corona homomorphism from G kills $\ker f$ and therefore factors uniquely through f. Conversely, composing a corona homomorphism of Q with f gives one of G. Intersecting these two corresponding families of kernels gives the displayed MF radical pullback. For the final assertion, functoriality of the MF radical under the inclusion $\ker f \to G$ gives $\ker f \le \mathrm{Rad}_{\mathrm{MF}}(G)$.

The same factorization, restricted to homomorphisms that kill N, gives the closure formula. If this closure equals N, then it contains $\ker f$. Mapping the equality onto Q shows that $\mathrm{cl}^{Q}_{\mathrm{MF}}(f(N)) = f(N)$. Conversely, these two conditions give

$$\mathrm{cl}^{G}_{\mathrm{MF}}(N) = f^{-1}(f(N)) = N.$$

The final equivalence now follows from Proposition 1.1, applied in G and Q. □

Thus image under f and inverse image under f give mutually inverse correspondences, preserving inclusion between the normal subgroups with MF quotient in G and in Q. The map induced by f on the largest quotients visible to MF is an isomorphism:

$$G/\mathrm{Rad}_{\mathrm{MF}}(G) \cong Q/\mathrm{Rad}_{\mathrm{MF}}(Q).$$

These correspondences respect composition of surjective homomorphisms with kernels invisible to MF.

Let B be a countable group with $\mathrm{Rad}_{\mathrm{MF}}(B) = B$, and let $1 \ne d \in B$ normally generate B. Put $A = \langle d \rangle$. For a countable group Q, define

$$W_Q = B *_A (Q \times A), \tag{16}$$

where $A \to Q \times A$ is the inclusion into the second factor. The trivial map $B \to Q$ and the projection $Q \times A \to Q$ agree on A. They therefore define an epimorphism

$$\pi_Q \colon W_Q \longrightarrow Q.$$

It is split by the inclusion $Q \to Q \times A \to W_Q$. The normal-form theorem for amalgamated free products shows that both vertex maps in (16) are injective; in particular, $d \ne 1$ in W_Q.

Proposition 7.2 (universal factorization)**.** *Let* T *be a group such that every homomorphism* $B \to T$ *is trivial. Then precomposition with* π_Q *is a bijection*

$$\mathrm{Hom}(Q, T) \longrightarrow \mathrm{Hom}(W_Q, T).$$

Furthermore,

$$\ker \pi_Q = \langle\!\langle d \rangle\!\rangle_{W_Q}.$$

Proof. Let $f\colon W_Q \to T$ be a homomorphism. Its restriction to B is trivial, so its restriction to the amalgamated subgroup A is trivial. For $(q,a) \in Q \times A$, we then have

$$f(q,a) = f(q,1)f(1,a) = f(q,1).$$

Thus the restriction of f to $Q \times A$ factors uniquely through the projection onto Q. The universal property of the amalgam shows that f factors through π_Q. The factorization is unique because π_Q is surjective.

The element d belongs to $\ker \pi_Q$. Conversely, quotienting W_Q by $\langle\langle d\rangle\rangle_{W_Q}$ kills B, since $\langle\langle d\rangle\rangle_B = B$, and kills the A-factor of $Q \times A$. The resulting quotient is Q, with quotient map induced by π_Q. Hence $\ker \pi_Q = \langle\langle d\rangle\rangle_{W_Q}$. □

Proof of Theorem C. Every homomorphism from B to an MF group is trivial: compose it with a faithful corona embedding of the target and use $\mathrm{Rad}_{\mathrm{MF}}(B) = B$. Proposition 7.2 therefore gives the asserted bijection for every MF group M. In addition, the definition of $\mathrm{Rad}_{\mathrm{MF}}(B) = B$ says directly that every homomorphism from B to $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is trivial. Applying Proposition 7.2 with $T = \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ shows that every corona homomorphism from W_Q factors uniquely through π_Q. Intersecting their kernels gives

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \pi_Q^{-1}(\mathrm{Rad}_{\mathrm{MF}}(Q)).$$

The kernel of π_Q belongs to this radical, and it contains the nonidentity element d. Thus W_Q is non-MF. If Q is MF, its MF radical is trivial, and the displayed equality becomes

$$\mathrm{Rad}_{\mathrm{MF}}(W_Q) = \ker \pi_Q = \langle\langle d\rangle\rangle_{W_Q}.$$

□

If $N \trianglelefteq W_Q$, the same factorization gives

$$\mathrm{cl}^{W_Q}_{\mathrm{MF}}(N) = \pi_Q^{-1}\left(\mathrm{cl}^{Q}_{\mathrm{MF}}(\pi_Q(N))\right). \tag{17}$$

Indeed, a homomorphism from W_Q to an MF group kills N exactly when its unique factor through Q kills $\pi_Q(N)$. Proposition 1.1 and (17) give

$$W_Q/N \text{ is MF} \quad\Longleftrightarrow\quad \ker \pi_Q \le N \text{ and } Q/\pi_Q(N) \text{ is MF}.$$

Here, if $\ker \pi_Q \le N$, then $N = \pi_Q^{-1}(\pi_Q(N))$.

Taking $Q = \mathbb{Z}$ makes the quotient criterion especially concrete: for every normal subgroup $N \trianglelefteq W_{\mathbb{Z}}$,

$$W_{\mathbb{Z}}/N \text{ is MF} \quad\Longleftrightarrow\quad d \in N. \tag{18}$$

Indeed, every quotient of $\mathbb{Z}$ is MF, while $\ker \pi_{\mathbb{Z}} = \langle\langle d\rangle\rangle_{W_{\mathbb{Z}}}$.

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Claude Fable 5 and GPT-5.6 Sol were used extensively in mathematical exploration, proof development, manuscript drafting, literature navigation, Lean formalization, and editorial revision. The Lean development verifies the one-sided transport theorem and the normal property-(T) radical criterion. It also verifies the rank 12 compression and commutator calculations and the resulting full MF radical conclusion. For Proposition 6.3, the formalization proves that every nontrivial normal subgroup contains an elementary matrix $e_{ij}(x)$ with $x \neq 0$. It treats diagonal matrices separately and divides the nondiagonal case according to whether the relevant entry of the inverse matrix is zero. The formalized simplicity proof uses the concrete two-sided sandwich and the fact that every central unit of the binary Leavitt algebra over $\mathbb{F}_2$ equals 1. It does not formalize Preusser's general theorem about normal subgroups or a general theory of pure infiniteness.

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